SCENTINEL

OBP candidate selection · analysis report

iGEM URV 2025

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Target VOC       :  methyl salicylate
CAS              :  119-36-8
Top N candidates :  10
Interferents     :  - none -
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tau (selectivity):  5
w1 affinity      :  0.450
w2 selectivity   :  0.250
w4 stability     :  0.150
w5 promiscuity   :  0.150
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SMAA configuration

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Method 2 (Perturbation):  c = 50%,  1,000 iterations
Method 3 (pure SMAA)   :  1,000 iterations
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SCENTINEL · Detailed results | VOC: methyl salicylate

#	OBP	Conserv.	Conf.	c2	s1	s2	s4	s5	Ki (uM)	Interf	Danger	Type
1	DcitOBP3	0.4919	0.74	0.00	0.472	0.000	1.000	0.902	7.01	-	-	Classic OBP
2	AcerOBP11	0.4833	0.74	0.00	0.479	0.000	1.000	0.814	6.83	-	-	Classic OBP
3	AcerOBP2	0.4657	0.73	0.00	0.427	0.000	1.000	0.888	8.29	-	-	Classic OBP
4	GmolOBP7	0.4632	0.75	0.00	0.381	0.000	1.000	0.964	9.81	-	-	Classic OBP
5	GmolPBP2	0.4452	0.74	0.00	0.452	0.000	0.850	0.793	7.55	-	-	PBP
6	PxylGOBP2	0.4399	0.74	0.00	0.437	0.000	0.850	0.795	7.97	-	-	GOBP2
7	SmosOBP12	0.4361	0.74	0.00	0.386	0.000	1.000	0.780	9.62	-	-	Classic OBP
8	RpadOBP3	0.4351	0.72	0.00	0.350	0.000	1.000	0.951	11.00	-	-	Classic OBP
9	RpadOBP7	0.4087	0.72	0.00	0.294	0.000	1.000	0.935	13.50	-	-	Classic OBP
10	HoblOBP2	0.4045	0.75	0.00	0.248	0.000	1.000	0.970	16.04	-	-	Classic OBP

Conserv. = Score x Confidence · c2 = selectivity confidence · Interf = measured/total · Danger = vineyard interferents binding as strong as the target
Row tint: green = confidence >= 0.80 lilac = 0.50-0.80 pink = < 0.50

Dangerous vineyard interferents per candidate

(Ki_interf <= Ki_target x 1.33; v = binds stronger than the target)

OK No candidate in the top has measured vineyard interferents binding as strong as the target.

SCENTINEL · s5 superbinders | VOC: methyl salicylate

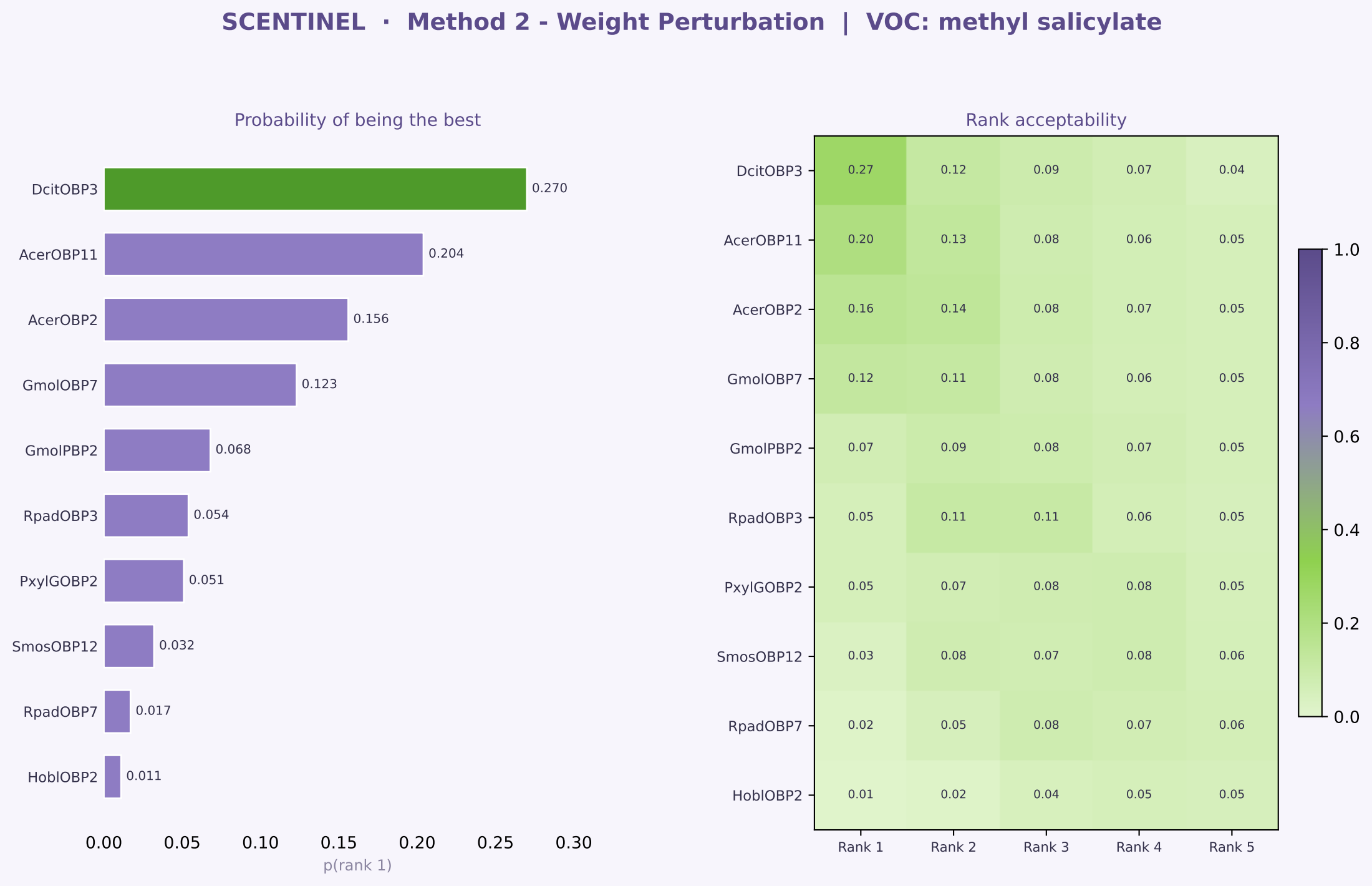
Superbinders - VOCs (any, not only vineyard) bound strongly

* = Ki <= 2.0 uM (strong) v = binds stronger than the target [n] = total superbinders

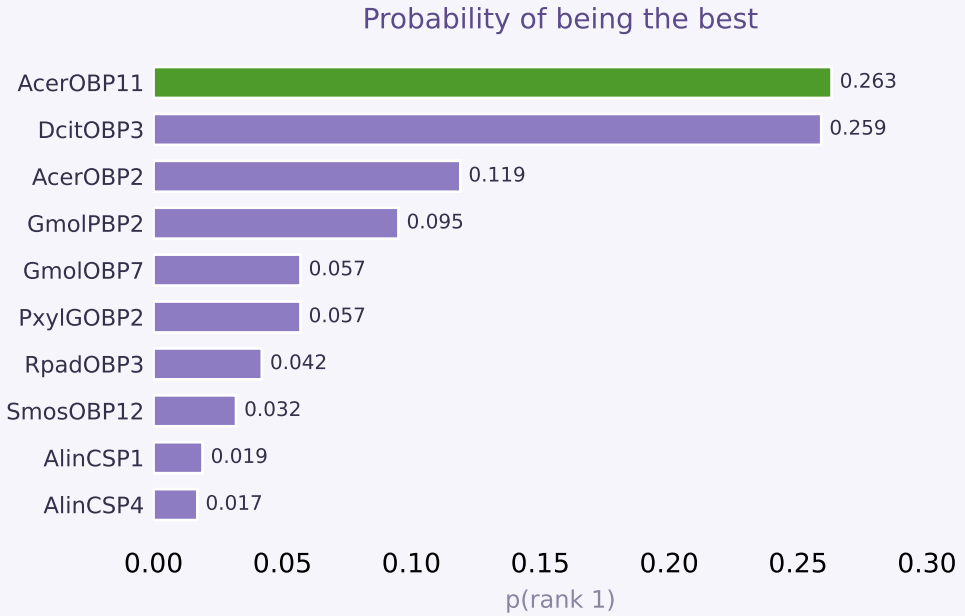
1.	DcitOBP3	[11]	DL linalool(3.5v), dimethyl trisulfide(4.2v), C9:Ald(4.3v), p-cymene(4.7v), (1R)-(+)-alpha-pinene(5.0v), (1S)-(-)-alpha-pinene(5.1v), dimethyl disulfide(5.1v), acetic acid(5.3v), trans-(3E)-beta-ocimene(5.9v), lauric acid(6.4v) +1
2.	AcerOBP11	[16]	HOB(1.4*v), Methyleugenol(2.7v), 1-hexanol(2.8v), isoamyl acetate(3.3v), phenylethanal(3.8v), ionone (beta)(3.8v), Ethyl hexadecanoate(3.8v), eucalyptol(4.3v), 2-heptanone(4.3v), l-(-)-Menthol(4.3v) +6
3.	AcerOBP2	[4]	phenylethanal(2.7v), ionone (beta)(4.1v), geraniol(5.5v), HOB(7.6v)
4.	GmolOBP7	[4]	Pear ester(2.5v), 12:Ald(4.3v), 12:OH(7.5v), alpha-ocimene(7.8v)
5.	GmolPBP2	[8]	Z8-12:Ac(1.1*v), E8-12:AC(1.1*v), Z8-12:OH(2.0*v), alpha-pinene(2.0v), 12:OH(2.8v), heptanal(3.8v), Cis-3-Hexenyl acetate(5.8v), ionone (beta)(6.7v)
6.	PxylGOBP2	[17]	Farnesene(1.0*v), (Z)-3-Hexenyl isovalerate(2.4v), Z11-16:Ac(2.4v), caryophyllene(2.4v), Z11-16:OH(2.7v), Z11-16:Ald(2.9v), phenylethanal(4.2v), DL linalool(4.5v), benzoic acid(4.9v), geraniol(5.5v) +7
7.	SmosOBP12	[16]	3-hexanol(3.3v), heptyl acetate(3.3v), 3-Carene(3.8v), D-panthenol(4.1v), Pentyl pentanoate(4.2v), 1-undecanol(4.4v), 1-octen-3-ol(5.3v), propanoic acid, hexyl ester(5.9v), alpha-pinene(6.2v), 3-hexanone(6.3v) +6
8.	RpadOBP3	[4]	beta-farnesene(8.9v), Z-3-hexen-1-ol(9.2v), (1S)-(-)-alpha-pinene(9.7v), r-limonene(10.5v)
9.	RpadOBP7	[1]	beta-farnesene(6.4v)
10.	HoblOBP2	[4]	ionone (beta)(5.4v), Hexyl benzoate(8.0v), ionone (alpha)(10.4v), Butyl benzoate(16.0v)

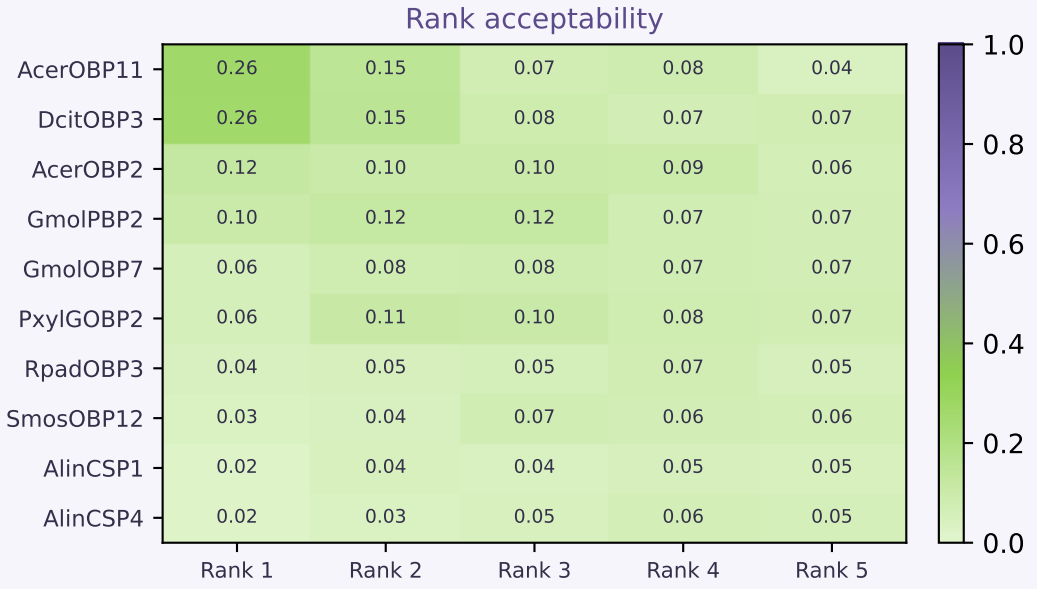
SCENTINEL · Experimental vs Computational | VOC: methyl salicylate

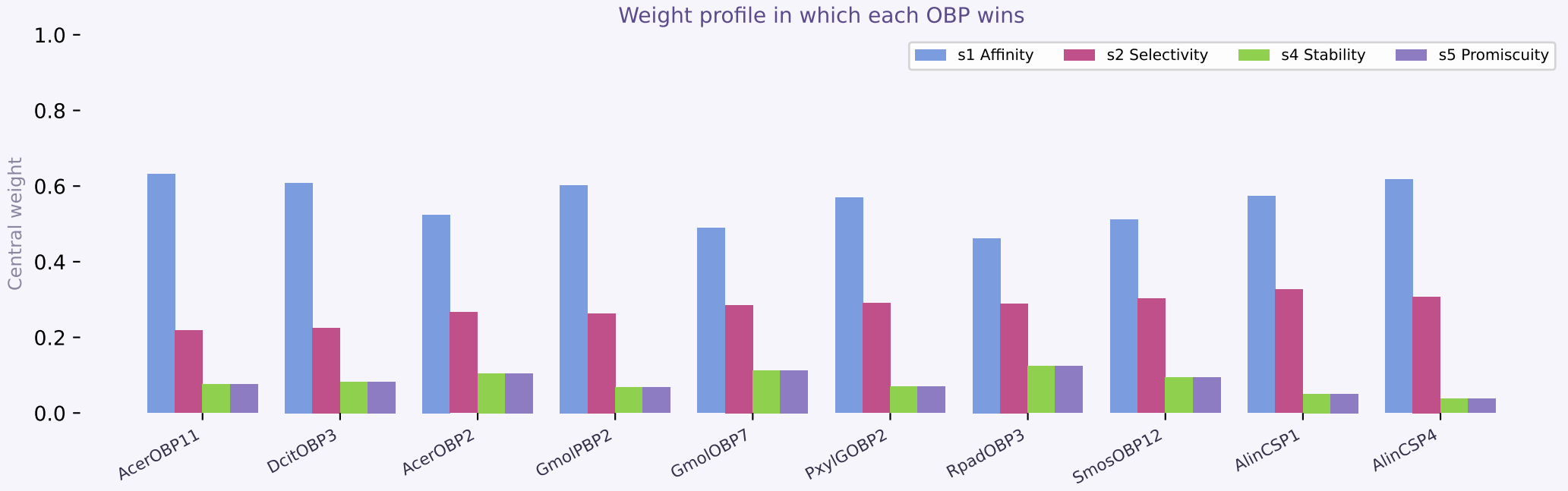



SCENTINEL · Method 2 - Weight Perturbation | VOC: methyl salicylate
Probability of being the best
DcitOBP3
AcerOBP11
AcerOBP2
GmolOBP7
GmolPBP2
RpadOBP3
PxylGOBP2
SmosOBP12
RpadOBP7
HoblOBP2
0.270
0.204
0.156
0.123
0.068
0.054
0.051
0.032
0.017
0.011
0.00
0.05
0.10
0.15
0.20
0.25
0.30
p(rank 1)
Rank acceptability
DcitOBP3 0.27 0.12 0.09 0.07 0.04
AcerOBP11 0.20 0.13 0.08 0.06 0.05
AcerOBP2 0.16 0.14 0.08 0.07 0.05
GmolOBP7 0.12 0.11 0.08 0.06 0.05
GmolPBP2 0.07 0.09 0.08 0.07 0.05
RpadOBP3 0.05 0.11 0.11 0.06 0.05
PxylGOBP2 0.05 0.07 0.08 0.08 0.05
SmosOBP12 0.03 0.08 0.07 0.08 0.06
RpadOBP7 0.02 0.05 0.08 0.07 0.06
HoblOBP2 0.01 0.02 0.04 0.05 0.05
Rank 1
Rank 2
Rank 3
Rank 4
Rank 5
1.0
0.8
0.6
0.4
0.2
0.0

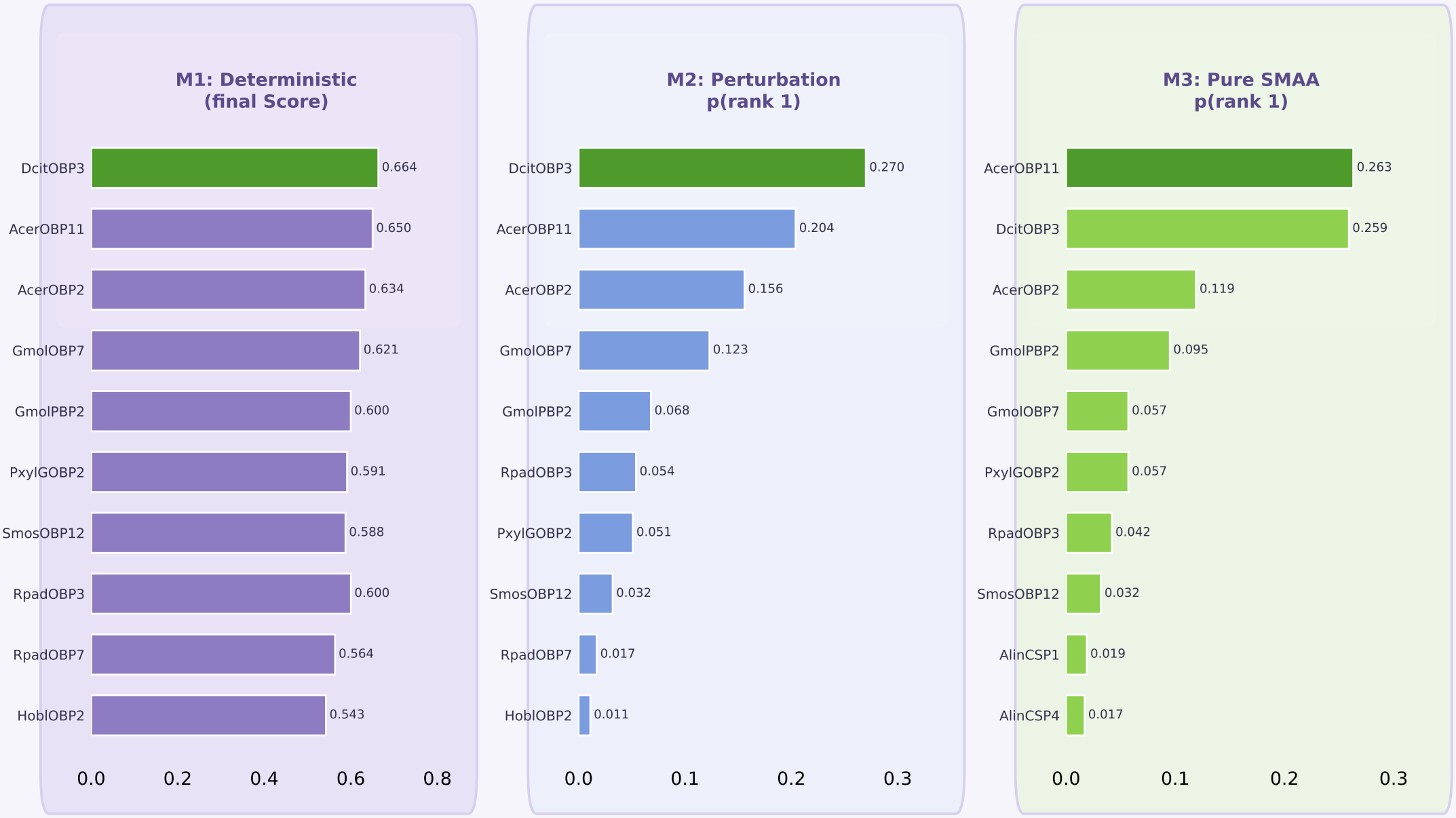
SCENTINEL · Method 3 - Pure SMAA | VOC: methyl salicylate







SCENTINEL · Comparison of the 3 methods | VOC: methyl salicylate



Winner · M1 (deterministic): DcitOBP3 M2 (perturbation): DcitOBP3 M3 (pure SMAA): AcerOBP11